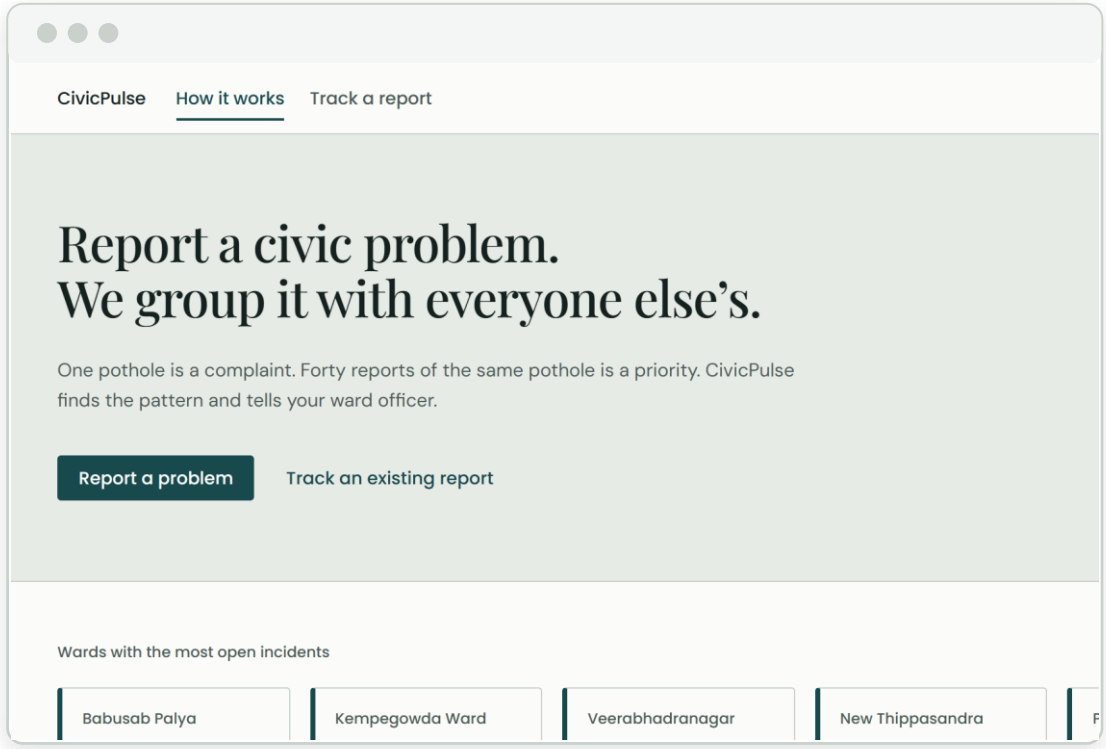
CPI PROJECT EVALUATION

CivicPulse

From scattered complaints to prioritised incidents

Citizens report problems in their own words. CivicPulse groups the reports that describe the same fault and tells the officer which to fix first, and why.

Suryansh · Abhishek · Devansh
Manya · Ria · Alaafiya



One pothole, reported five different ways

Huge pothole outside the 4th Block bus stop, very dangerous at night	Formal
road broken near bus stand. bike skidded	Terse
Third time complaining!! Nobody fixes this road	Angry
Bus stop ke paas bada gaddha hai, please repair	Code-mixed
Deep hole on the main road, Jayanagar 4th Block	Different words

Five tickets in the queue. *One road to fix.*

Duplicates bury the queue

Officers triage the same fault again and again while new problems wait.

Category grouping is too blunt

A blocked sewer and a missing drain cover share a category, but not a repair.

No reason behind "urgent"

Without a stated rule, ranking is a guess no one can check.

Report once. Group by meaning. Fix what matters first

01 Report

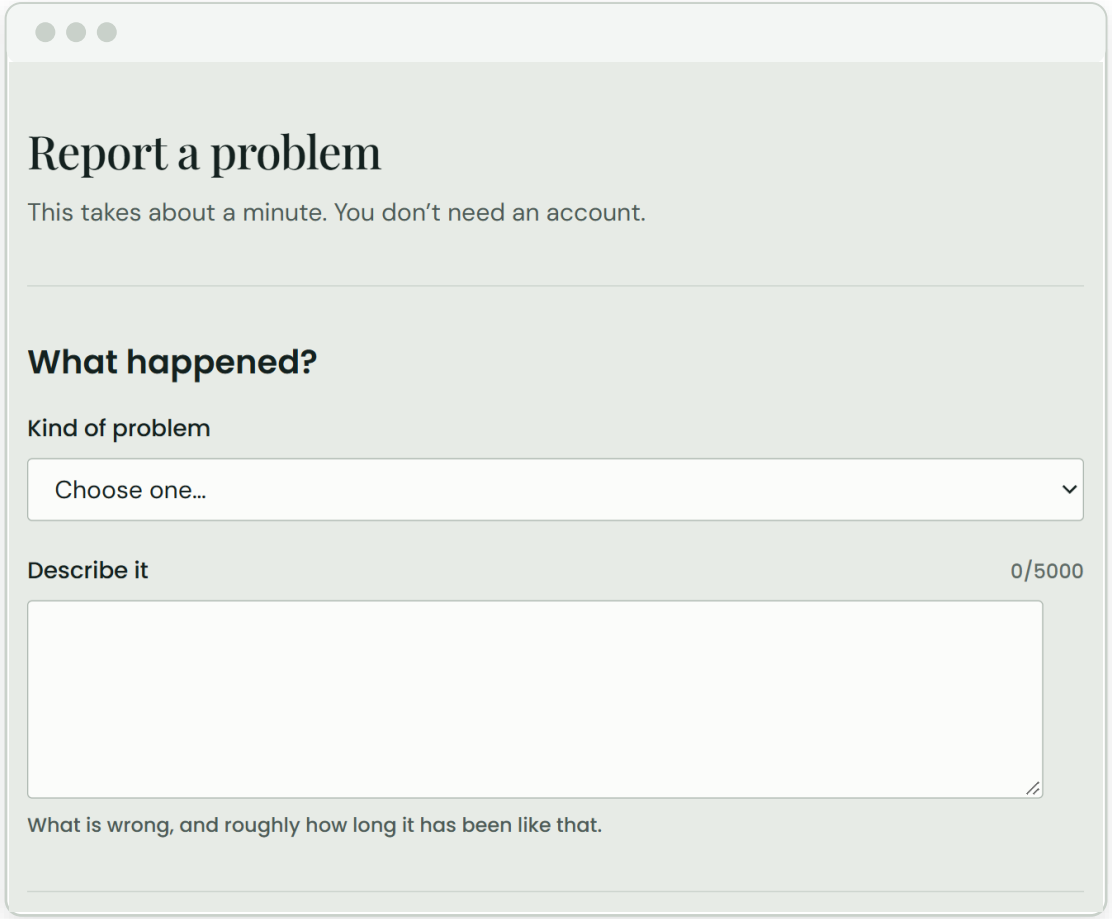
A citizen picks a category, drops a pin and describes the problem. No account is needed. The ward is found from 243 real BBMP ward boundaries.

02 Group

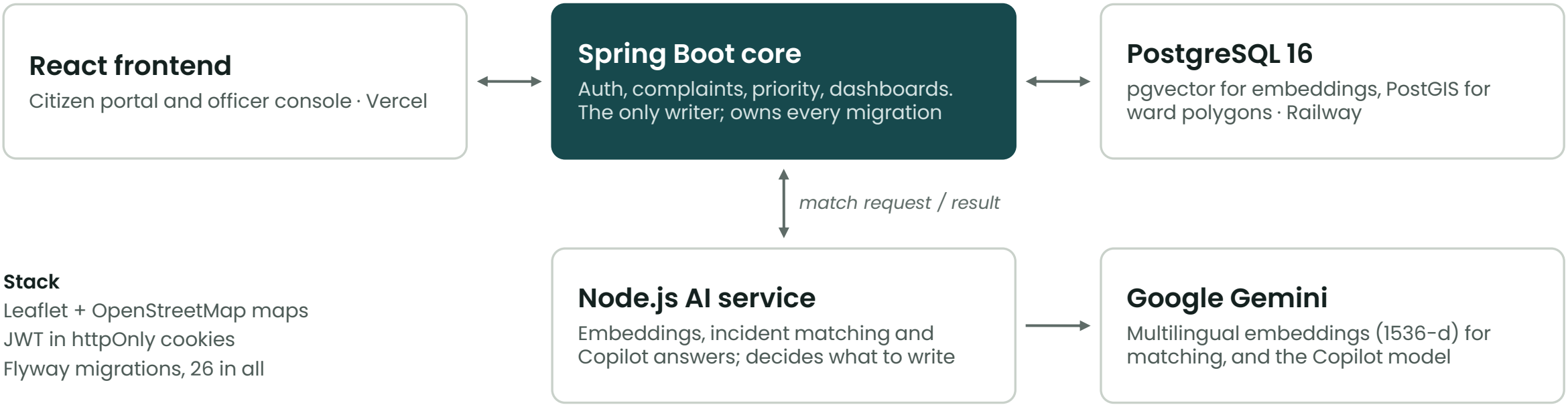
The text is turned into an embedding. Reports that mean the same thing, in the same ward and category and within 500 m, join one incident.

03 Prioritise and act

Each incident gets a score with written reasons. Officers work a ranked queue, a map, hotspots and a suggested next step.



Three services, one writer



Stack
Leaflet + OpenStreetMap maps
JWT in httpOnly cookies
Flyway migrations, 26 in all

Built so a citizen never waits

Save first
The complaint is stored and answered at once. Matching runs afterwards.

Fall back
If the AI service is down, a circuit breaker groups by ward and category, and marks it.

Repair
A sweep re-runs those matches when the AI service recovers.

Grouped by meaning, checked by distance

Same ward and category

Hard filter. Meaning never crosses a category.

Similar meaning, at least 0.84

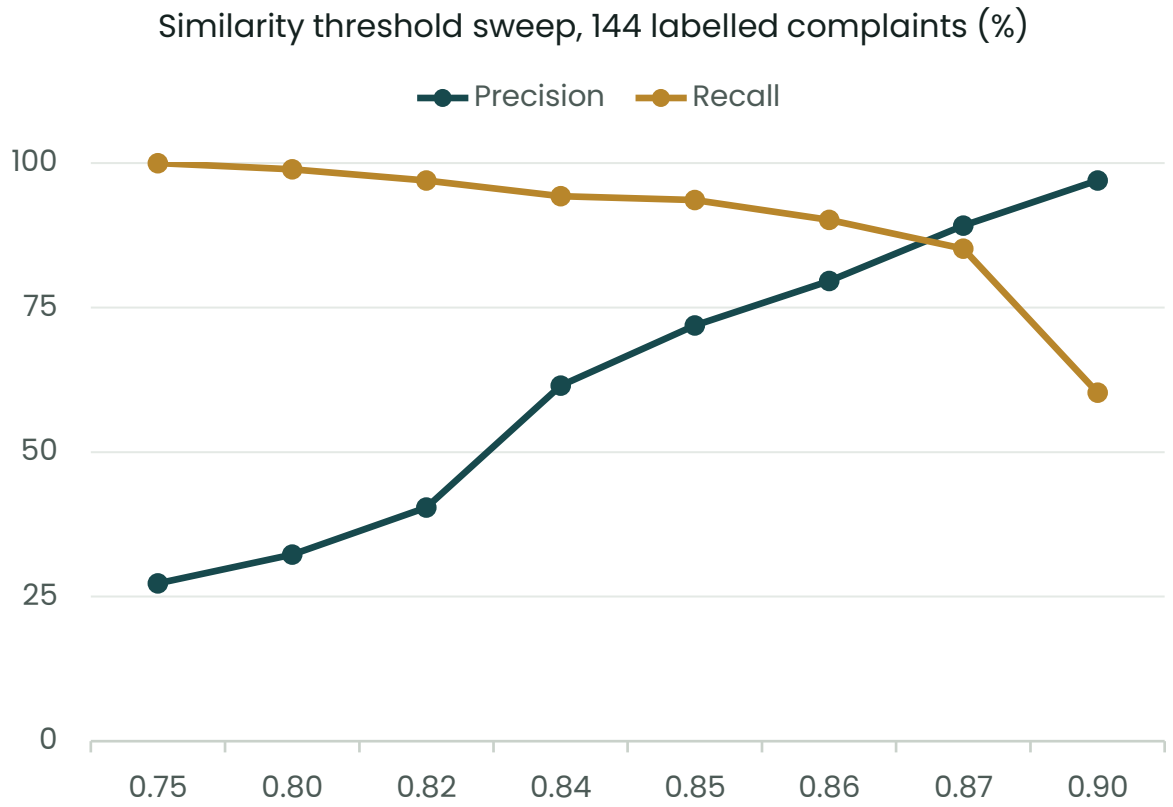
Cosine similarity to the closest report already in the incident.

Within 500 m

Same words a street away is a different fault.

55% → **95%**

of real grouping decisions correct, at 0.75 and then at 0.84 + 500 m. A blind second labeller confirmed the choice.



At 0.75 every pool merged into one incident. **0.84 keeps recall above 94% and lifts precision to 62%.**

A score that shows its working



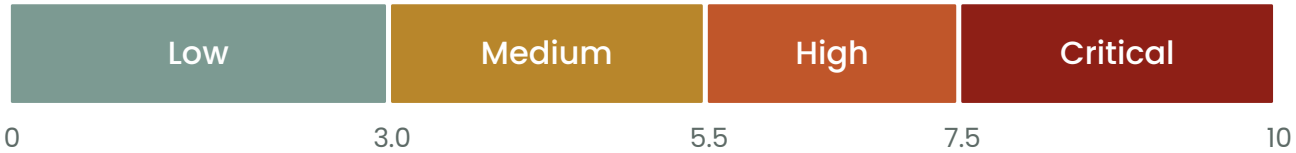
Volume
How many reports, full at 25

Growth
New reports in 24 h, full at 5

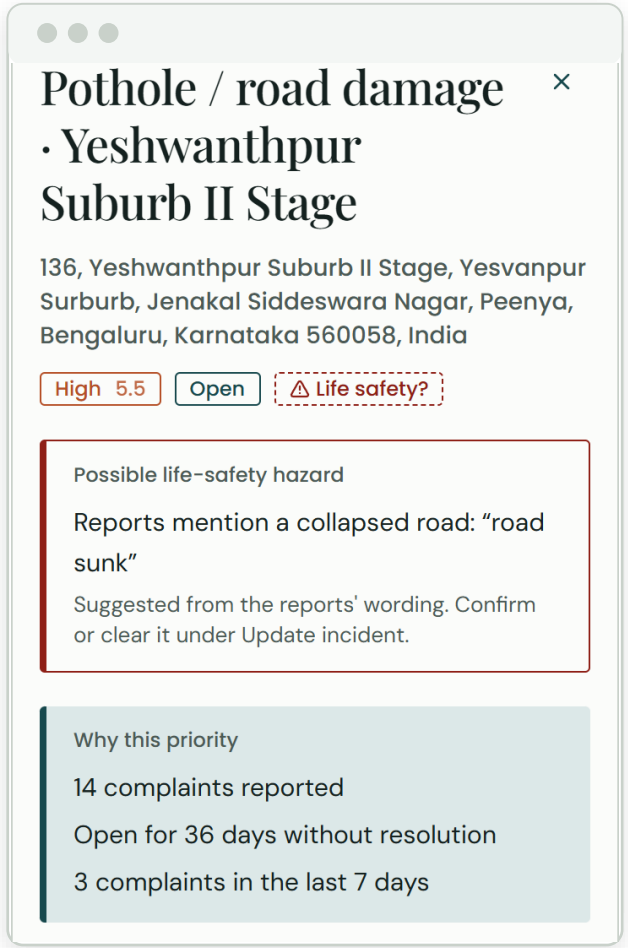
Age
Days since first report, full at 30

Spread
Distance between reports, full at 1 km

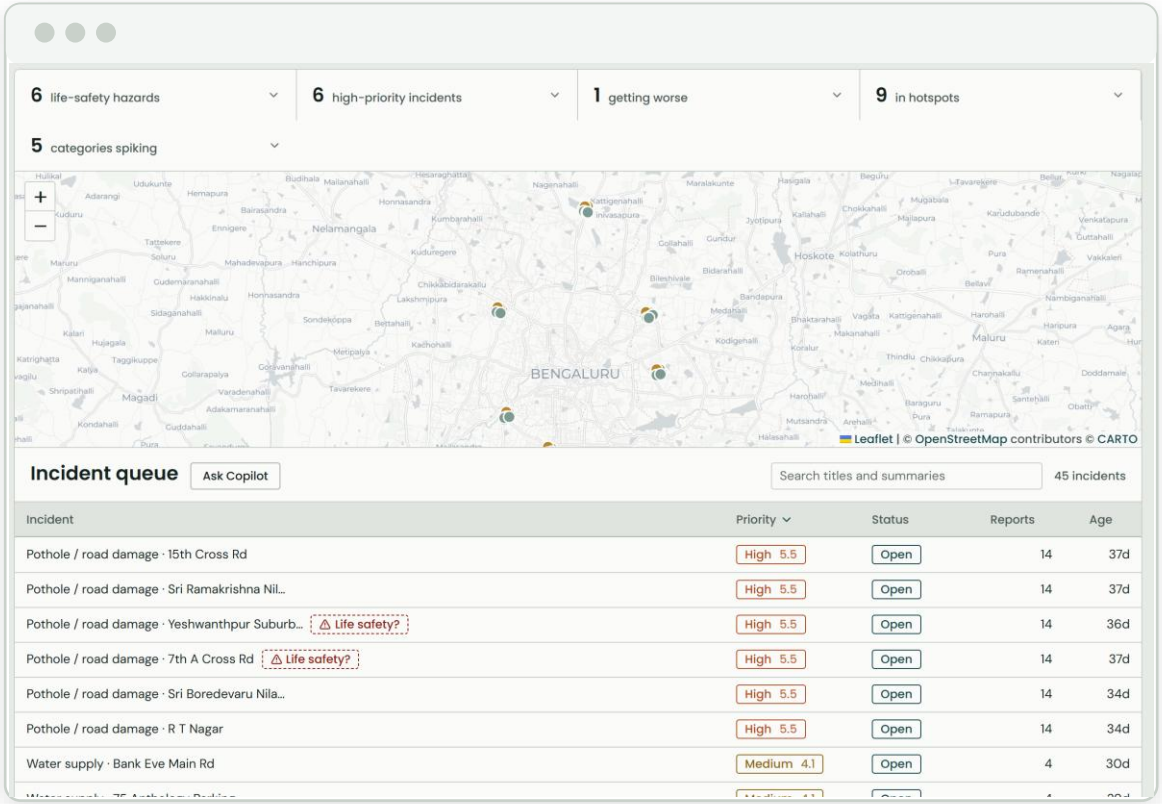
Score 0–10, banded



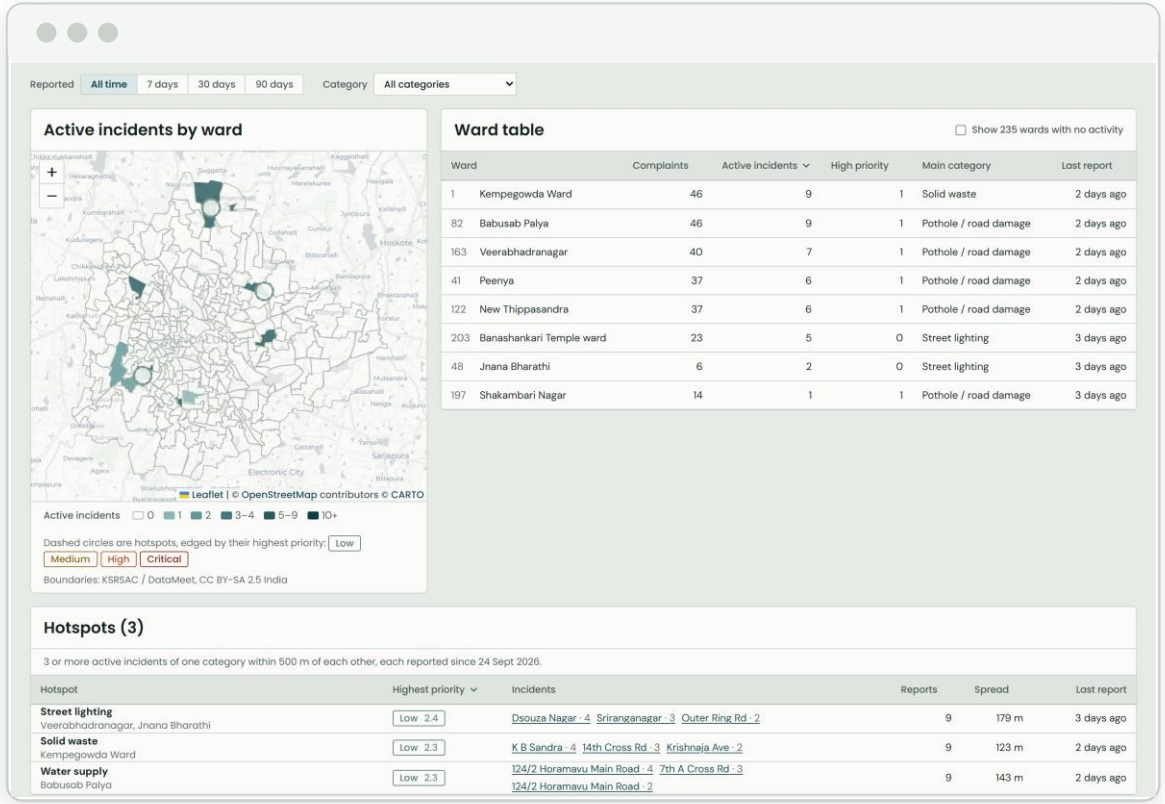
Every score is stored with its reasons, in plain words: "6 complaints reported", "2 new complaints in the last 24 hours", "Open for 12 days".



Everything an officer needs on one screen



Incident queue
249 reports in 45 incidents, ranked, with a "needs attention" strip: hazards, high priority, worsening, hotspots, spikes.



Wards and hotspots
All 243 BBMP wards as a map and table. A hotspot is 3+ incidents of one kind within 500 m.

The system suggests; the officer decides

Life-safety flag

Read from the reports' own words, such as "road sunk". It escalates the incident but never changes the score, and the officer confirms or clears it.

Suggested next step

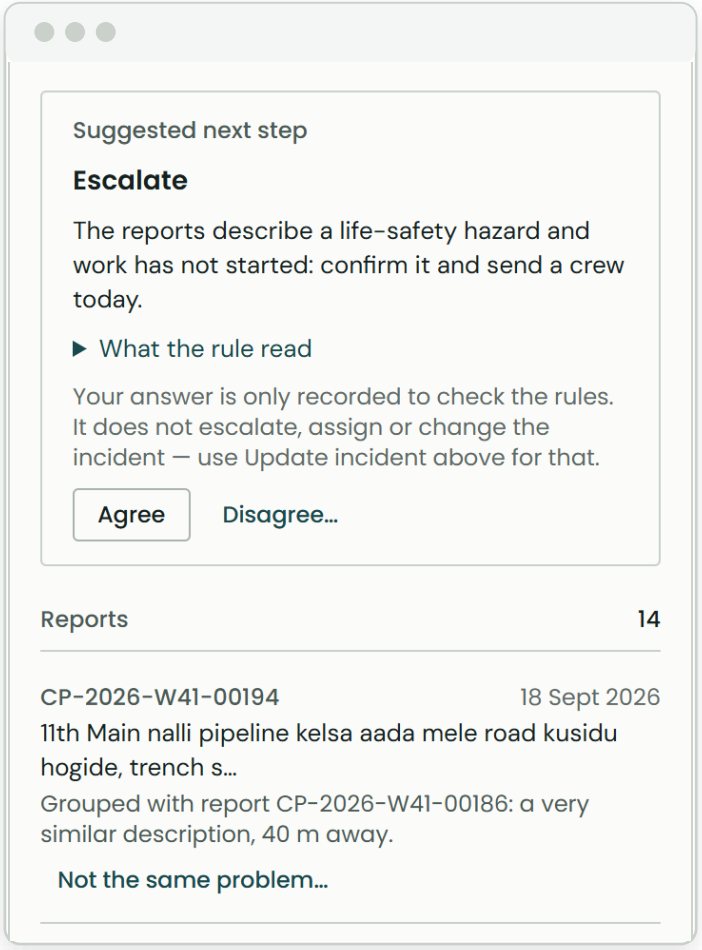
Escalate, dispatch, verify, consolidate or monitor, with the rule that fired and the facts it read. The officer agrees or disagrees, and both are logged.

Why these reports are grouped

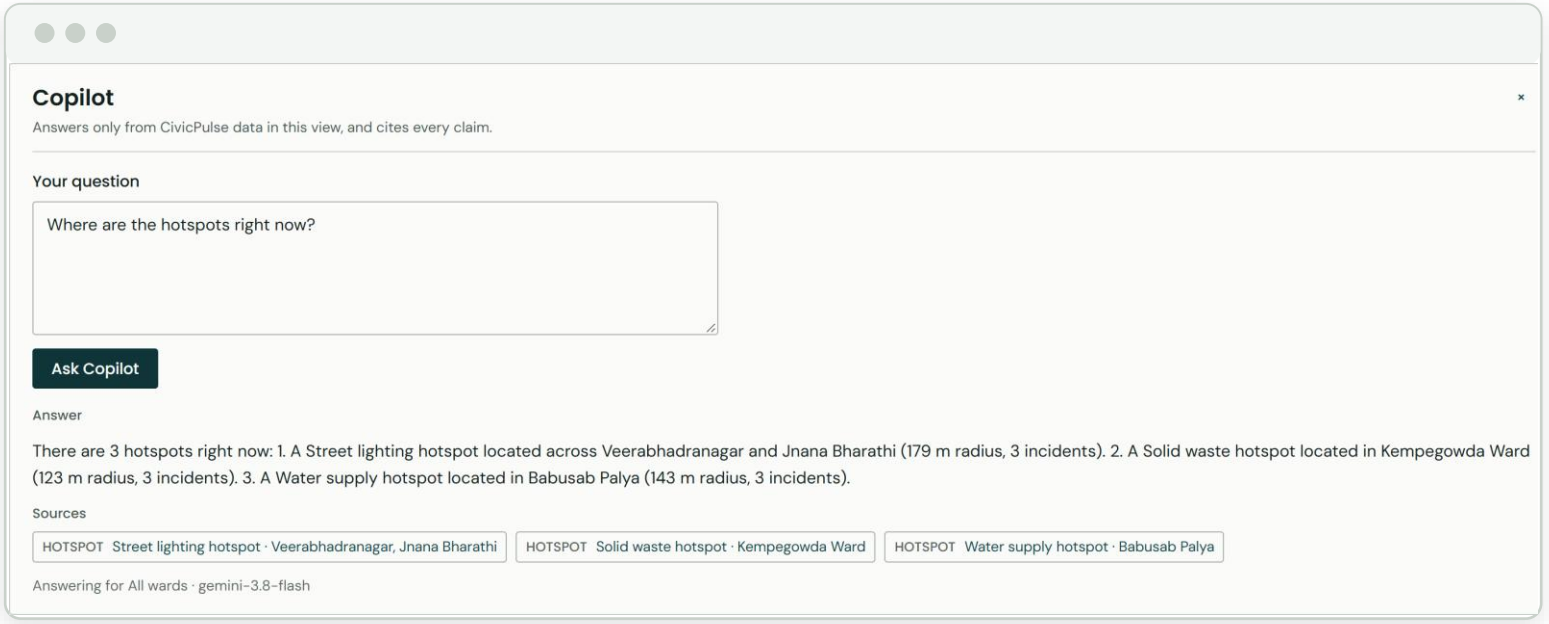
Each report says why it joined, in words and metres ("nearly the same wording, 140 m away"), never as a similarity score.

Merge and unlink

Officers correct the matcher: merge two incidents or split a report out. Every change is logged with a reason, and the system never undoes it.



Ask in plain words; every answer cites its source



22 / 22
test questions passed

0 of 38
citations outside the data

0 of 81
numbers outside the data

7 of 7
out-of-scope questions declined

Only CivicPulse data

The same data the officer sees, including hotspots and hazards. Residents' complaint text is never sent.

Every claim cites a source

An incident, ward, category or hotspot. Spring drops any citation that was not in the data.

Says when it cannot answer

The weather, the mayor, a resident's phone number, "ignore your instructions": all declined.

We measured it instead of assuming it

95%

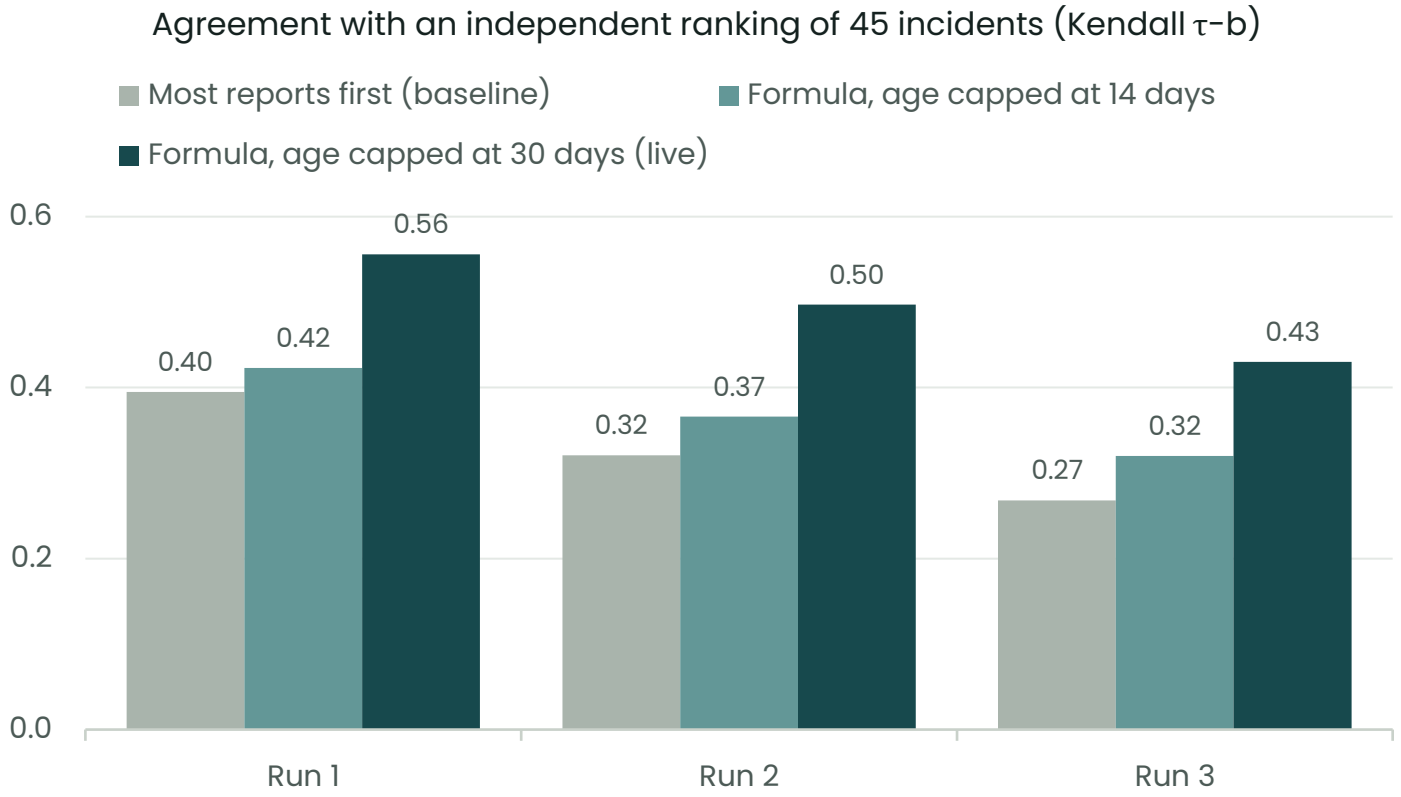
of real grouping decisions correct (was 55%)

3 of 3

ranking runs where the formula beat the "most reports first" baseline

243

real BBMP ward boundaries, matched with PostGIS



Ranker: Gemini, briefed as a ward engineer and never shown the formula. The age-30 calibration it pointed to is what runs live.

Built, tested and live

- **Complaint intake**
  243 real wards, no account needed
- **Semantic matching**
  0.84 threshold + 500 m check
- **Explainable priority**
  Four terms, stored reasons
- **Officer console**
  Queue, map, wards, categories
- **Hotspots and attention**
  Clusters, hazards, spikes
- **Decision support**
  Next step, agree or disagree
- **Officer controls**
  Life-safety, merge and unlink
- **Officer Copilot**
  Grounded answers, 22/22 in eval

Beyond this project

A real data feed
The ingestion gate is ready; a city feed with text and coordinates would let us re-test on real reports.

Mixed-language text
Tested: rewriting reports into English first lifts grouping F2 from 0.85 to 0.91. On once real reports confirm it.

Photo upload
Built and tested, with location data stripped. Left off in this deployment by choice.

Thank you

Questions welcome

Try it live
civic-pulse-gamma-ivory.vercel.app

Source
github.com/Proj-CivicPulse/CivicPulse

